Project Initialization and Planning Phase

Date	27-june-2026
Team ID	SWUID2026223535
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Proposed Solution Template:

S.No.	Parameter	Description
1.	**Problem Statement (Problem to be solved)**	Banks receive thousands of credit card applications every day. Manual verification is time-consuming and may lead to errors. The project aims to automate the credit card approval process using Machine Learning for faster and more accurate decisions.
2.	**Idea / Solution description**	A Flask-based web application is developed that predicts whether a credit card application should be approved or rejected based on applicant details. Multiple ML algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost are trained, and the best-performing model is used for prediction.
3.	**Novelty / Uniqueness**	The system compares multiple machine learning models and selects the best one for prediction. It provides instant results through a user-friendly web interface and supports cloud deployment for real-time access.
4.	**Social Impact / Customer Satisfaction**	The solution reduces manual effort, speeds up the approval process, minimizes human errors, improves customer experience, and helps banks make consistent and reliable credit decisions.
5.	**Business Model (Revenue Model)**	Banks and financial institutions can subscribe to the application as a Software-as-a-Service (SaaS) platform. Revenue can also be generated through enterprise licensing, API integration, and annual maintenance services.
6.	**Scalability of the Solution**	The application can be deployed on cloud platforms like IBM Cloud and can handle thousands of credit card applications in real time. It can also be integrated with existing banking systems and expanded with new datasets and advanced machine learning models.